1 Introduction and Model Set-up

This paper is about testing the validity of a linear factor asset pricing model using two-pass regression method.

Here we take the view that the β's of an asset is an important/fundamental property of the asset, so instead of considering where β are potentially nonlinear functions of conditioning variables, i.e., $\beta = \beta(z)$, or considering the time varying risk premium, which still specifies that the expected return is a linear function of β, we test whether the expected return is a nonlinear function of β.

Suppose there are k-dimenisonal observed candidate factors $F_t := (F_{1,t}, \dots, F_{k,t})$ for $t = 1, \dots, T$. Let Y_{it} be the excess returns. We want to test

$$H_0 : E(Y_i \mid X_i) = X_i'\gamma \quad a.s. \quad \text{for some} \quad \gamma \in \Gamma \tag{1}$$

$$H_1 : P\big(E(Y_i \mid X_i) = X_i'\gamma\big) < 1 \quad \text{for all} \quad \gamma \tag{2}$$

where $X_{ik} = \operatorname{cov}(Y_i, F_k)/\operatorname{var}(F_k)$ and we assume the time-series dynamic is

$$Y_{it} = X_i' F_t + \varepsilon_{it} \tag{3}$$

where ε_{it} are assume to be indepepdent across t and independent of F_t.

Assumption 1 (Model under H_0)**.** *In addition to the model in Equation 3, we make the following assumptions on the data generating process*

1. *β_i are independent across i and $\beta = \left[\beta_1, \dots, \beta_N\right]'$ is independent of F_t for all t.*
2. *$\varepsilon_t = (\varepsilon_{1t}, \dots, \varepsilon_{Nt})'$ are independent over time t. $E\varepsilon_t\varepsilon_t' = \Sigma$, there may be cross-sectional correlation. Assume that $\|\Sigma\| < \infty$.*
3. *F_t are stationary and $E(\varepsilon_t \mid \beta, F_t) = 0$.*

2 Lemmas on Error of Local Linear Estimator

2.1 Estimation Error With Local Linear Estimator

Following Shanken, 1992, if H_0 is true and Equation 3 holds, let $m_0(x, \gamma) = x'\gamma$, because

$$E(Y_i) = X_i' E(F) \quad \bar{Y}_i = X_i' \bar{F} + \bar{\varepsilon}_i \tag{4}$$

$$E(Y_i) = m_0(X_i, \gamma) \quad \bar{Y}_i = \bar{m}_0(X_i, \gamma) + \bar{\varepsilon}_i \tag{5}$$

we have

$$\bar{m}_0(X_i, \gamma) = m_0(X_i, \gamma) + X_i' \left(\bar{F} - E(F)\right)$$

Let $\bar{M} = \begin{bmatrix} \bar{m}_0(X_1, \gamma) \\ \vdots \\ \bar{m}_0(X_n, \gamma) \end{bmatrix}$ and $\bar{\varepsilon} = \begin{bmatrix} \bar{\varepsilon}_1 \\ \vdots \\ \bar{\varepsilon}_n \end{bmatrix}$.

For any variable a_{it}, $\bar{a}_i = \frac{1}{T}\sum_t a_{it}$. In the first step we use OLS,

$$\hat{X}_i = (F'F)^{-1} F' Y_i \quad \text{so} \quad \hat{X} = Y_{(n\times T)} F \left(F'F\right)^{-1}$$

For $x \in \operatorname{supp}(X) \subset \mathbb{R}^k$, let local linear estimator be $\hat{m}(x) = e_1^\top \left(\hat{X}_x' \hat{W}_x \hat{X}_x\right)^{-1} \hat{X}_x' \hat{W}_x \bar{Y}$ where $e_1^\top = (1, 0, \dots, 0)'$

$$\hat{X}_x = \begin{bmatrix} \mathbf{1} & \left(\hat{X} - \mathbf{1} \otimes x'\right) \end{bmatrix} \quad \hat{W}_x = \operatorname{diag}\left(K_h\left(\hat{X}_1 - x\right), \dots, K_h\left(\hat{X}_n - x\right)\right)$$

Let $\theta(x) = (m(x), Dm(x))'$.

Assumption 2. *We restrict the alternative by assuming $m(x) \in \mathcal{C}^2$. Let $K(\cdot)$ be a symmetric probability density function with compact support. $K_h(\cdot) = \frac{1}{h} K\left(\frac{\cdot}{h}\right)$.*

Then we have

$$\begin{aligned}
\hat{\theta}(x) - \theta(x) &= \left(\hat{X}_x' \hat{W}_x \hat{X}_x\right)^{-1} \hat{X}_x' \hat{W}_x \bar{Y} \\
&= \hat{S}_x^{-1} \hat{X}_x' \hat{W}_x \left(\bar{M} + \bar{\varepsilon} - \hat{X}_x \theta(x)\right) \\
&= \hat{S}_x^{-1} \hat{X}_x' \hat{W}_x \left(M - X_x' \theta(x) + X(F_t - EF) + \left(\hat{X}_x - X_x\right)' \theta(x) + \bar{\varepsilon}\right)
\end{aligned}$$

e where $\hat{S}_x = \hat{X}_x' \hat{W}_x \hat{X}_x$.

Lemma 1. *We have the following decompositions:*

1. $\hat{X}_x = X_x + R_1$ *where* $R_1 = \begin{bmatrix} \mathbf{0} & \hat{X} - X \end{bmatrix} = O_p\left(\frac{1}{\sqrt{T}}\right)$;

2. $\hat{W}_x = W_x + R_2$, *where*

$$R_2 = \operatorname{diag}\left(\frac{1}{h^2}K'\left(\frac{X_1 - x}{h}\right)\left(\hat{X}_1 - X_1\right), \dots, \frac{1}{h^2}K'\left(\frac{X_n - x}{h}\right)\left(\hat{X}_n - X_n\right)\right) \tag{6}$$

$$= O_p\left(\frac{1}{h^2\sqrt{T}}\right) \tag{7}$$

3. $\frac{1}{n}\hat{S}_x = S_x + Remainders.$

Proof. $\hat{X}_x = \begin{bmatrix} \mathbf{1}_n & \hat{X} - \mathbf{1}_n x' \end{bmatrix} = \begin{bmatrix} \mathbf{1}_n & X - \mathbf{1}_n x' \end{bmatrix} + \begin{bmatrix} 0 & \hat{X} - X \end{bmatrix} = X_x + O_p\left(\frac{1}{\sqrt{T}}\right)$.

Assuming K_h has bounded support and bounded first-order derivative, then by Taylor expansion we have the estimate for $\hat{W}_x$. □

Lemma 2. *For* $k = 1$ *we will have*

$$\hat{X}_x' \hat{W}_x \hat{X}_x = \begin{bmatrix} \sum K_h\left(\hat{X}_j - x\right) & \sum K_h\left(\hat{X}_j - x\right)\left(\hat{X}_j - x\right) \\ * & \sum K_h\left(\hat{X}_j - x\right)\left(\hat{X}_j - x\right)^2 \end{bmatrix} \tag{8}$$

Normalize by $1/n$ *we have*

$$\frac{1}{n}\hat{S}_x = \begin{bmatrix} f(x) + O_p(h^2 + \frac{1}{\sqrt{nh}} + \frac{1}{h^2\sqrt{T}}) & * \\ * & \dagger \end{bmatrix} \tag{9}$$

The $*$ *part is*

$$* \quad = o_p(1) + O_p\left(\frac{1}{\sqrt{T}}\right) + O_p\left(\frac{1}{h\sqrt{T}}\right) + O_p\left(\frac{1}{h^2 T}\right) \tag{10}$$

The $\dagger$ *part is*

$$\dagger = h^2 \kappa_2 f(x) + remainder \tag{11}$$

2.2 Specification Test

We show that this fits into the framework of Escanciano, Pardo-Fernández, and Keilegom, 2018. We calculate the following quantities:

$$\hat{\varepsilon}_{j,0} = \frac{Y_j - m_0\left(X_j, \hat{\gamma}\right)}{\hat{\sigma}(X_j)} \quad \text{and} \quad \hat{\varepsilon}_j = \frac{Y_j - \hat{m}\left(X_j\right)}{\hat{\sigma}(X_j)}$$

References

Escanciano, Juan Carlos, Juan Carlos Pardo-Fernández, and Ingrid Van Keilegom (June 2018). "Asymptotic Distribution-Free Tests for Semiparametric Regressions with Dependent Data". In: *Annals of Statistics* 46.3, pp. 1167–1196. ISSN: 0090-5364, 2168-8966. DOI: 10.1214/17-AOS1581. URL: https://projecteuclid.org/euclid.aos/1525313079 (visited on 06/17/2020).

Shanken, Jay (Jan. 1, 1992). "On the Estimation of Beta-Pricing Models". In: *The Review of Financial Studies* 5.1, pp. 1–33. ISSN: 0893-9454. DOI: 10.1093/rfs/5.1.1. URL: https://academic.oup.com/rfs/article/5/1/1/1599649 (visited on 02/22/2020).